a.Simulation and Performance Evaluation of People–Process–Data–Things IoE Framework using Edge/Fog-to-Cloud Pipeline with MQTT, Node-RED.

b.Benchmarking of 5G/6G Communication for IoE Applications based on Throughput, Latency, Reliability and Energy Efficiency

Major: IoE Framework and Emerging Networks

Minor: People–Process–Data–Things, Edge/Fog, 5G/6G

PROGRAM CODE:

OBJECTIVE A:

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# EXPERIMENT 6(a): People-Process-Data-Things (P-P-D-T) & Edge/Fog/Cloud Pipeline

# Output Checked: Complete P-P-D-T data flow; edge/fog/cloud performance

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import numpy as np

import pandas as pd

import matplotlib.pyplot as plt


np.random.seed(42)

n_cycles = 40

times = pd.date_range("2026-10-31 11:00:00", periods=n_cycles, freq="1S")

pressure_psi = np.random.normal(101.0, 6.0, n_cycles)

pressure_psi[12:17] += 24.0  # Surge event


ppdt_records = []

for i in range(n_cycles):

    # 1. THINGS

    p_val = round(float(pressure_psi[i]), 2)

    # 2. DATA (Edge/Fog/Cloud Latency & Bandwidth reduction)

    edge_lat = round(float(np.random.normal(3.5, 0.4)), 2)
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    fog_lat = round(float(np.random.normal(11.8, 1.1)), 2)
    cloud_lat = round(float(np.random.normal(38.0, 3.5)), 2)

    # 3. PROCESS & 4. PEOPLE
    if p_val > 118.0:
        proc = "ACTUATE_RELIEF_VALVE"
        person = "Safety_Engineer (SMS Alert)"
    elif p_val > 108.0:
        proc = "THROTTLE_PUMP_SPEED"
        person = "Shift_Operator (Dashboard Warn)"
    else:
        proc = "ROUTINE_ARCHIVE"
        person = "Automated_System (No Action)"

    ppdt_records.append({
        "Time": times[i].strftime("%H:%M:%S"),
        "Things_PSI": p_val,
        "Data_Edge_ms": edge_lat,
        "Data_Fog_ms": fog_lat,
        "Data_Cloud_ms": cloud_lat,
        "Process_Decision": proc,
        "People_Notified": person
    })

df_6a = pd.DataFrame(ppdt_records)
print("=" * 85)
print("EXPERIMENT 6(a): COMPLETE P-P-D-T DATA FLOW LOGS")
print("=" * 85)
```

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print(df_6a[["Time", "Things_PSI", "Data_Edge_ms", "Process_Decision",
"People_Notified"]].iloc[10:18].to_string(index=False))

print("=" * 85)


fig, axes = plt.subplots(1, 3, figsize=(15, 4.5))


# 1. Things & Process Tracking

axes[0].plot(df_6a["Time"], df_6a["Things_PSI"], color="#2980b9", lw=2)

axes[0].axhline(118.0, color="#c0392b", linestyle="--", label="P1 Relief Limit")

axes[0].axhline(108.0, color="#f39c12", linestyle=":", label="P2 Warning Limit")

axes[0].set_title("Things & Process: Pressure Pipeline", fontweight="bold")

axes[0].set_ylabel("Pressure (PSI)")

axes[0].tick_params(axis="x", rotation=45)

axes[0].legend()

axes[0].grid(True, linestyle="--", alpha=0.5)


# 2. Edge / Fog / Cloud Performance Comparison

tiers = ["Edge Tier", "Fog Tier", "Cloud Tier"]

latencies = [df_6a["Data_Edge_ms"].mean(), df_6a["Data_Fog_ms"].mean(),
df_6a["Data_Cloud_ms"].mean()]

axes[1].bar(tiers, latencies, color=["#27ae60", "#f39c12", "#8e44ad"], width=0.45)

axes[1].set_title("Data Layer: Edge vs Fog vs Cloud Latency", fontweight="bold")

axes[1].set_ylabel("Average Response Latency (ms)")

axes[1].grid(axis="y", linestyle="--", alpha=0.5)


# 3. People Notification Distribution

p_counts = df_6a["People_Notified"].value_counts()

axes[2].bar(p_counts.index, p_counts.values, color=["#2ecc71", "#e67e22", "#e74c3c"], width=0.45)

axes[2].set_title("People Layer: Responder Dispatch Count", fontweight="bold")
```

```
axes[2].tick_params(axis="x", rotation=18)

axes[2].set_ylabel("Event Count")

axes[2].grid(axis="y", linestyle="--", alpha=0.5)


plt.suptitle("Experiment 6(a): P-P-D-T IoE Framework & Edge/Fog/Cloud Performance",
fontsize=13, fontweight="bold")

plt.tight_layout()

plt.show()
```

OBJECTIVE B:

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# EXPERIMENT 6(b): Benchmarking of 5G/6G Communication for IoE Applications

# Output Checked: 5G/6G comparison graphs for throughput, latency, reliability, energy

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import pandas as pd

import matplotlib.pyplot as plt


bench_df = pd.DataFrame({

    "Benchmark Metric": [

        "Peak Throughput (Gbps)",

        "User Plane Latency (ms)",

        "Reliability / Error Rate (PER)",

        "Energy Efficiency (pJ/bit)"

    ],

    "5G (IMT-2020)": [20.0, 1.0, 1e-5, 100.0],

    "6G (IMT-2030)": [1000.0, 0.1, 1e-7, 10.0]

})
```

```
print("=" * 75)
print("EXPERIMENT 6(b): 5G vs 6G BENCHMARKING RESULTS")
print("=" * 75)
print(bench_df.to_string(index=False))
print("=" * 75)

fig, axes = plt.subplots(2, 2, figsize=(11, 7))

axes[0, 0].bar(["5G", "6G"], [20.0, 1000.0], color=["#2980b9", "#27ae60"], width=0.4)
axes[0, 0].set_yscale("log")
axes[0, 0].set_title("Throughput Comparison (Gbps - Log Scale)", fontweight="bold")
axes[0, 0].set_ylabel("Gbps")
axes[0, 0].grid(True, linestyle="--", alpha=0.5)

axes[0, 1].bar(["5G", "6G"], [1.0, 0.1], color=["#e67e22", "#c0392b"], width=0.4)
axes[0, 1].set_title("Latency Comparison (ms)", fontweight="bold")
axes[0, 1].set_ylabel("Latency (ms)")
axes[0, 1].grid(axis="y", linestyle="--", alpha=0.5)

axes[1, 0].bar(["5G", "6G"], [1e-5, 1e-7], color=["#8e44ad", "#16a085"], width=0.4)
axes[1, 0].set_yscale("log")
axes[1, 0].set_title("Reliability (Packet Error Rate - Log Scale)", fontweight="bold")
axes[1, 0].set_ylabel("Packet Loss Probability")
axes[1, 0].grid(True, linestyle="--", alpha=0.5)

axes[1, 1].bar(["5G", "6G"], [100.0, 10.0], color=["#d35400", "#2c3e50"], width=0.4)
axes[1, 1].set_title("Energy Efficiency Comparison (pJ/bit)", fontweight="bold")
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axes[1, 1].set_ylabel("Energy per Bit (pJ/bit)")

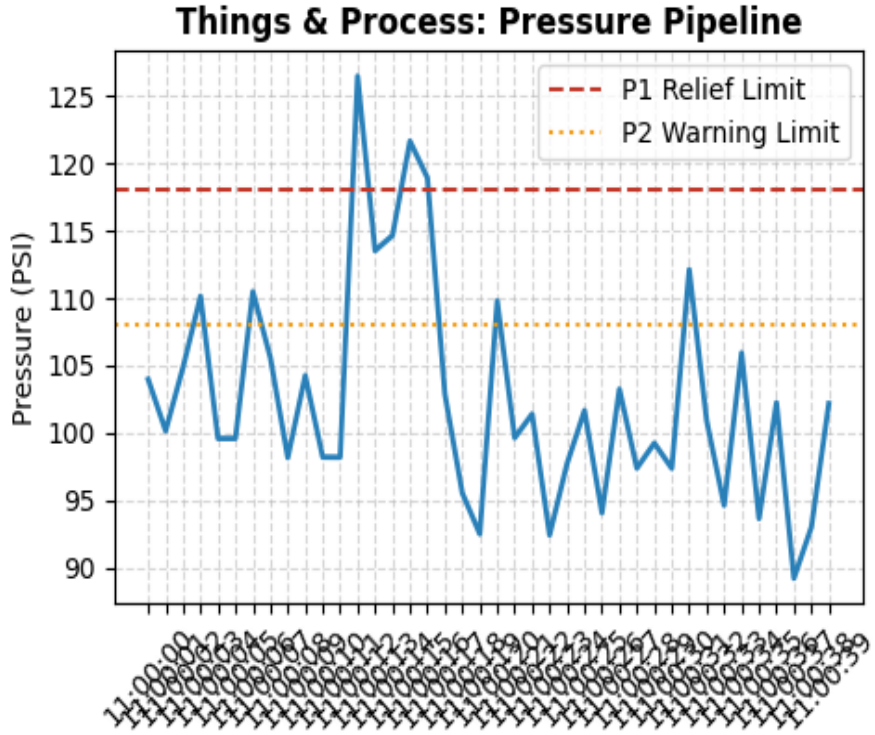
axes[1, 1].grid(axis="y", linestyle="--", alpha=0.5)

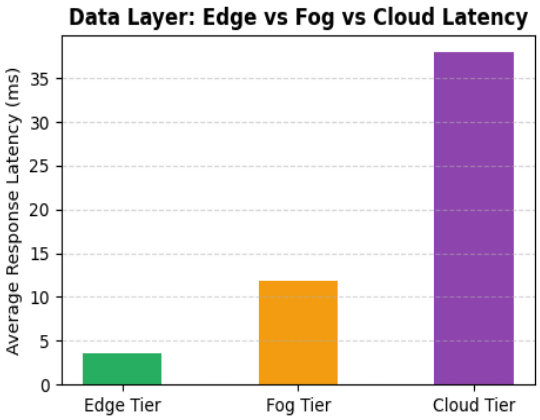

plt.suptitle("Experiment 6(b): 5G vs 6G Communication Benchmarking for IoE", fontsize=13,
fontweight="bold")

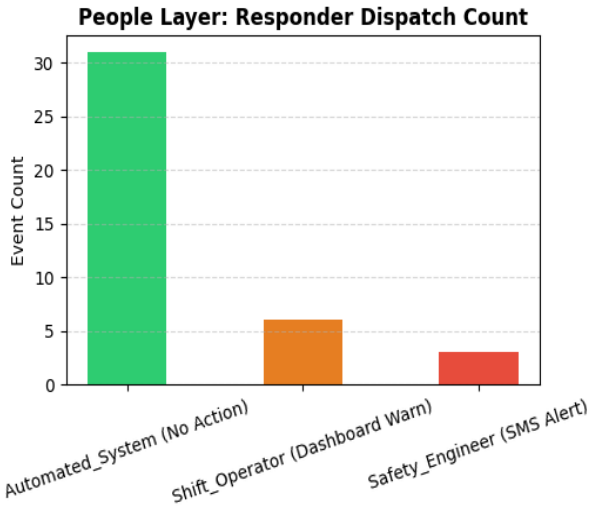
plt.tight_layout()

plt.show()
```

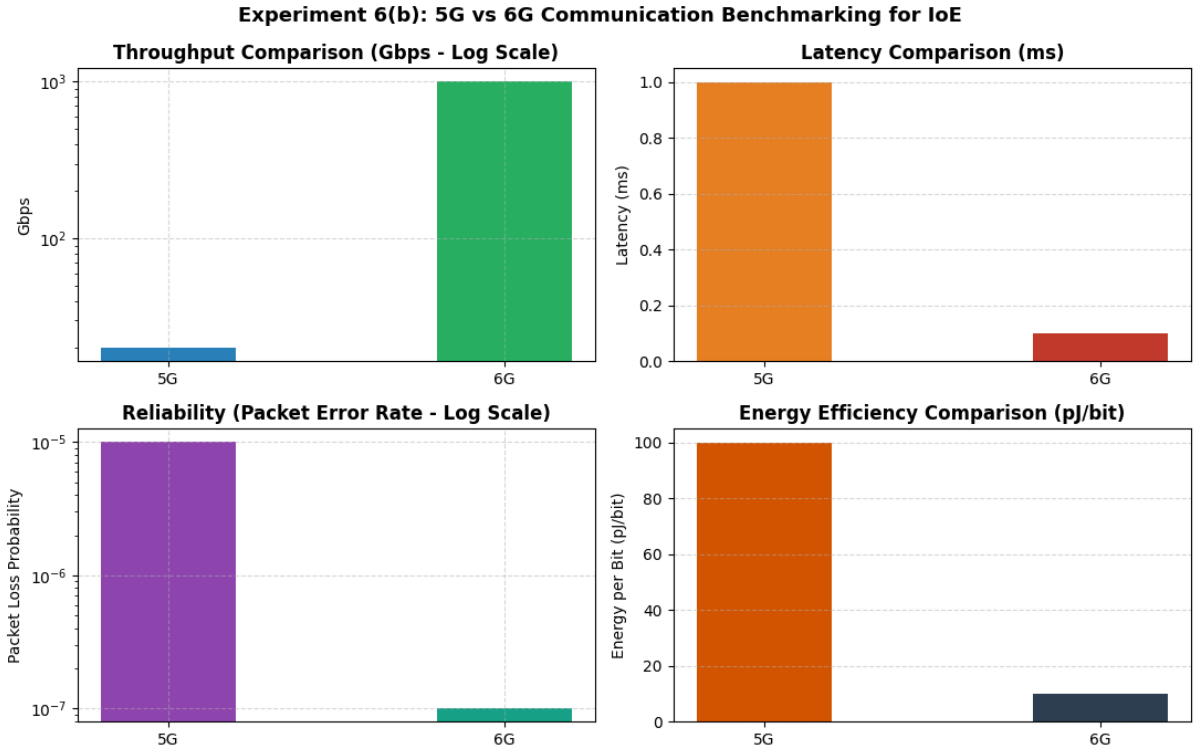
OUTPUT IMAGE:
OBJECTIVE A:







OBJECTIVE B:



RESULT:

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EXPERIMENT 6(a): COMPLETE P-P-D-T DATA FLOW LOGS

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Time	Things_PSI	Data_Edge_ms	Process_Decision	People_Notified
11:00:10	98.22	3.64	ROUTINE_ARCHIVE	Automated_System (No Action)
11:00:11	98.21	4.13	ROUTINE_ARCHIVE	Automated_System (No Action)
11:00:12	126.45	3.53	ACTUATE_RELIEF_VALVE	Safety_Engineer (SMS Alert)
11:00:13	113.52	2.70	THROTTLE_PUMP_SPEED	Shift_Operator (Dashboard Warn)

11:00:14 114.65 4.09 THROTTLE_PUMP_SPEED Shift_Operator (Dashboard Warn)

11:00:15 121.63 3.30 ACTUATE_RELIEF_VALVE Safety_Engineer (SMS Alert)

11:00:16 118.92 3.29 ACTUATE_RELIEF_VALVE Safety_Engineer (SMS Alert)

11:00:17 102.89 3.89 ROUTINE_ARCHIVE Automated_System (No Action)

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EXPERIMENT 6(b): 5G vs 6G BENCHMARKING RESULTS

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Benchmark Metric	5G (IMT-2020)	6G (IMT-2030)
Peak Throughput (Gbps)	20.00000	1.000000e+03
User Plane Latency (ms)	1.00000	1.000000e-01
Reliability / Error Rate (PER)	0.00001	1.000000e-07
Energy Efficiency (pJ/bit)	100.00000	1.000000e+01

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